Data Literacy and Research Data Management: The Case at ULSIT

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Abstract. In 2017 a team of researchers from the University of Library Studies and Information Technologies (ULSIT) joined an international research group to conduct a scientific survey on Data Literacy and Research Data Management. In the research paper we examine the current state of data literacy, data sharing and reuse perceptions and practices among researchers at ULSIT. The research, which found gaps in understanding of data management best practices, provides recommendations for data literacy training to be provided in the near future. The implementation of the survey was timely, necessary, and useful in light of current amendments to the Act for the Development of the Academic Staff in the Republic of Bulgaria (2018). Analysis of the current levels of awareness and gaps in knowledge among the survey population has resulted in the creation of a *document with recommendations* for changes to the university's data management policy that also includes specific improvements for the university's Quality of Education Management System.

Keywords: Research data management · Data literacy · Open access · Higher education · Training · ULSIT

1 Introduction

Within the frame of the 2016 European Conference on Information Literacy (ECIL), researchers from the United Kingdom, France and Turkey initiated an international research project on Data Literacy (DL) and Research Data Management (RDM) [1]. The core research group developed a questionnaire which aimed to collect data about the data literacy of faculty and doctoral students in higher education institutions. According to Schneider [2] DL or Research Data Literacy (RDL) are dimensions of information literacy that are dedicated to the creation, management, and reuse of research data. RDL is defined as a subfield of RDM, which is concerned with processing of all types of raw or primary data that are created along every research process.

In Bulgaria, the need of implementation of the RDM policy is connected with open data policy implementation, underlined by two national strategies - "National Strategy for Development of Scientific Research in the Republic of Bulgaria 2017–2030: Better Science for Better Bulgaria" (NSDSR) and "Bulgaria National Roadmap for Research Infrastructure, 2017–2023" [3, 4]. This created new demands on researchers, who are faced with a complex of challenges and skills requirements that they have not dealt with previously. These skills are crucial to ensuring data quality, integrity, shareability,

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discoverability, and reuse over time [5]. Convinced that the survey on Data Literacy (DL) and Research Data Management (RDM) was both timely and necessary, a team of researchers from the University of Library Studies and Information Technologies (ULSIT) joined the international scientific group and conducted the research.

2 The Aim and Methodology

The questionnaire survey "Data Literacy Survey" (DLS) contains two groups of questions. The first group collects respondent demographic information, and the second aims to determine competence regarding management of research data and the kind of training that is needed for researchers in the context of RDM. In order to find answers to these questions, an online questionnaire survey was conducted amongst university faculty, research staff, and doctoral students at ULSIT. The implementation of the survey at ULSIT passed through three stages [6]. During the first stage in January 2017, a translation of the questionnaire instruments from English language into Bulgarian was carried out and a list of respondents was created. During the second stage, from February to March 2017, the online survey (LimeSurvey) was disseminated to participants and the data was gathered. The second stage, February–March 2017 includes dissemination of the participation invitation in the online based survey and accumulation of data. The third stage involved analysis of the results, creation of a summary document with conclusions and recommendations for consideration by stakeholders at ULSIT, and the writing of a scientific article. Participation in forums and events in Bulgaria and abroad has been planned that are aimed at continuing conversation on the significant range of problems identified by the authors.

The data obtained were subjected to logical verification and control and then processed through the statistical package SPSS (Statistical Package for the Social Sciences), version 19.0. For specification of the psychometric characteristics of the methods and verification of the hypotheses, other statistical processing methods were also applied: descriptive statistics, correlation analysis, single-factor dispersion analysis.

3 Findings

In ULSIT, the survey was conducted on the basis of systematic random sampling with stratification of 10% of all professors under basic employment contracts at the university and 10% of matriculating doctoral students. The general aggregation consists of 40 effectively surveyed lecturers and doctoral students, which makes the survey representative for ULSIT. From the target group of 147 researchers, recruited in ULSIT, 27.2% (n = 40) respondents answered to the whole survey.

3.1 Demographics

Survey responses totalled 40: 90% (n = 36) were lecturers and 10% (n = 4) were PhD students. Out of 40 respondents, 80% (n = 32) were female and 20% (n = 8) were

male. Most of the participants were ages 36–45 (42% (n = 17). This is followed by those in the age range of 46–55 and 56–65 (20% respectively, n = 16). 15% (n = 6) were between the age of 26–35 and 3% (n = 1) were over 65 years old. Distribution of the respondents by scientific specialty is the following: 67% (n = 27) were from Social Sciences (mainly Library and Information Sciences) and 33% (n = 13) were from Computer Sciences. Research experience of the respondents were as follows: 5–10 years (27%, n = 11), less than 5 years (25%, n = 10), more than 20 years (18%, n = 7), and between 11–15 and 16–20 years (15% respectively, n = 12).

3.2 Awareness of Data Management Issues

The original online survey includes 24 questions. Respondents were asked to indicate the file type of data that they normally use for their research. Preference for use of non-digital data (e.g., paper, film, slides, artifacts), standard office documents (e.g., text, spreadsheets, presentations) and Internet and web-based data (e.g., webpages, e-mails, blogs, social network data) was expressed by 90% (n = 36), 85.0% (n = 34) and 70% (n = 28) of respondents respectively. A statistically significant difference in the respondents' responses was established in relation to their age difference. The conclusion is that younger researchers of age groups: 26–35 year olds, 36–45 year olds and 46–55 year olds, use a greater variety of data file types than those over the age of 55. This is probably due to the better computer skills of the researchers in the three age groups mentioned.

A statistically significant difference was found between the responses of respondents with different research experience on the question about description of the volume of data that they use and produce from their research. Logical and regular is the observed tendency for researchers with longer experience in the relevant scientific field to use and produce the larger volumes of information. Table 1 illustrates the finding that the volume of data that respondents produced from their research is directly related to the volume of information used in the research process. For example, the average for researchers with less than 5 years of research experience is M (mean) = 1.60, while for researchers with 11–15 years of research experience is M = 1.67, and for those with more than 20 years is M = 1.86. Only scientists with more than 20 years of research experience use and create the largest volume of data – one terabyte.

Table 1. Volume of data generated as a result of the research activities (in units of measurement of information)

Factor (Research experience)	N	M	SD	F	Sig	Min.	Max.
<5 years	10	1,60	0,51	1,50	0,22	1	2
5–10 years	11	1,45	0,52			1	2
11–15 years	6	1,67	0,51			1	2
16–20 years	6	1,77	0,40			1	2
>20 years	7	1,86	0,69			1	3

According to data obtained from the question: *"How do you usually get the data for your research?"*, more than half of the respondents create new research data themselves. 60% (n = 24) and 50% (n = 20) used data from their own research team or scientific group at the university. The percentage of researchers relying on data from their own research network or personal/professional connections is also relatively high. The results suggest that respondents do not limit themselves to a single source but prefer a variety of well-known and reliable sources - 80% (n = 32). Asked *"How do you usually use data that you get from others/outside sources?"* respondents usually gave more than one response from the given options. This suggests that the use of data from others/outside sources depends on the specific situation in which they use the relevant data. Most often, however, respondents prefer to spend a lot of time and effort making data usable for a particular study 85% (n = 34). The survey found that the respondents most often create the following type of data associated with their research: standard office documents 100% (n = 40); images (JPEG, GIF, TIFF, PNG) 55% (n = 22); archived data (ZIP, RAR, ZAR) 45% (n = 18), as well as non-digitalized data (paper, films, slides, artifacts) 45% (n = 18). The lack of use of structured graphics (CAD, CAM, VRML) and Raw (machine-generated) data is probably due to respondents' research areas.

From the results of the question *"Where do you usually store the data you produce from your research?"* it becomes clear that the respondents rely mainly on their own devices 87.5% (n = 35) or tools that reside in the cloud 40.0% (n = 16). Survey results find that most of the researchers apply additional supportive information to their research data: administrative information (e.g., creator, date of creation, file name, access terms/restrictions) 77.5% (n = 31); discovery information (e.g., creator, funding body, project title, project ID, keywords) 62.5% (n = 25); and technical information and description of the data file, both of which undoubtedly help to better organize and make usable the data arrays on both proprietary devices and in repositories 32.5% (n = 13).

Respondents also were asked to indicate their opinion about collaboration with other researchers and sharing of research data. More than half of respondents 62.5% (n = 25) are willing to share data and collaborate with other researchers, both from their own institution and from other scientific organizations. This fact is undoubtedly important for the development of national and international research networks. Nearly half of the respondents 45% (n = 18) are more likely to provide their research data and developments freely. Asked if they have any concerns about sharing data with others, almost half of respondents 45% (n = 18) said that their research data should be freely available to anyone who is interested in it, that is, they have no concern with the public availability of their data. A finding that deserves attention is, however, the concern of more than one-third of the respondents about the lack of appropriate copyright policies and rights protection - 35% (n = 14).

The last group of questions refers to *specific documents and activities for managing research data at the university* and aim to establish the *level of awareness of the respondents*. The analysis of the results shows that the picture of the individual respondents' opinions is quite colorful. The respondents' responses vary across the full spectrum of possibilities. Respondents are not well informed about the existence of the Data Management Plan (DMP) at the university, nor are they aware of the

implementation of the plan. The fact that most of the respondents 65% (n = 26) believe that the DMP effectively helps scientists manage their research data is encouraging. Also encouraging from the perspective of the authors is that respondents believe in the value and importance of metadata training relating to data description. According to 82.5% (n = 33) of the surveyed researchers, such training would be useful for research data management.

In Bulgaria, standards for citing data are well known and used. Since 2011, the international standard ISO 690: 2010 "Information and documentation: Guidelines for bibliographic references and citations to information resources" has been used as the national standard for citing data [7]. In this regard, it is understandable and natural that 92.5% (n = 37) of the respondents believe it important to use a specific standard style for citing and referring to research data.

In the survey questionnaire, participants were asked how often they engage in specific data management practices. Each statement was required to be chosen on a 5-point Likert scale with the following options: *almost always*, *often, sometimes, rarely, never*. Because the academic community is heavily involved in research, respondents were expected to have high level of DL and RDM competencies. Therefore the *satisfactory level* is determined from answers *almost always* and *often*, presented in Table 2.

Table 2. Respondents answers *almost always/often* on responses of the question *"How often do you practice the following?"*

Responses	n	%
Using metadata standard for tagging your data	13	32.5
Use your own/in-house (your research team) tags and metadata	14	35.0
Using datasets that are tagged with standard metadata	12	30.0
Use file naming convention or standard	20	50.0
Having different versions of the same dataset(s)	16	40.0
Using systems/techniques for version control to easily recognize a specific version	13	32.5
Citing research data	27	67.5
Working with data that are generally in the public domain	34	85.0
Working with data that have restricted access	13	32.5

Results of the survey indicate a far from satisfactory level of data management practices, except on the issue of citing research data and using public domain data.

Respondents also were asked to indicate their opinions regarding other DL and RDM statements relating to data sharing. 80% (n = 32) of respondents strongly agree or agree that they are familiar with open access (OA) requirements. Nevertheless, only 1 respondent (2.5% (n = 1) strongly agreed with the statement *'I am comfortable and willing to share my research data with others'*, while 82.5% (n = 33) disagree or strongly disagree with that statement. 37.5% (n = 15) of respondents strongly agree or agree that they foresee no problems with sharing their research data and more than half

of the respondents 62.5% (n = 25) perceive that data ethics could be an issue when research data is shared with others. The high correlation between these last two statements (r = 0.68) indicates that in practice the researchers from ULSIT are reluctant to share their research data due to potential ethical and copyright problems. This is a finding that requires further attention.

As for the statement *'I would like to store my research datasets beyond the lifetime of the project'*, while 67.5% (n = 27) strongly agreed, 27.5% (n = 11) neither agree nor disagree and 5% (n = 2) disagree or strongly disagree. A high majority of respondents 95% (n = 38) agree that every university should have a Data Management Plan, as well as that universities should recommend and use a standard file naming system, 87.5% (n = 35). A slight majority believe that universities should have a prescribed set of metadata elements for uploading data into a repository, 67.5% (n = 27). The survey found that, according to most of the respondents, financial provision should not be the concern of the researchers themselves or of their teams, but rather the responsibility of the university, other research funding bodies, and/or a national authority. Most of the researchers, 67.5% (n = 27), believe that scientific data should be stored at the university or in an external repository for long-term preservation and access and 50% (n = 20) believe it should be provided free of charge.

The last two questions in the questionnaire aim to establish if *formal trainings have been conducted* on the various components of RDM and *whether respondents would like to be involved in future trainings*. From the data presented it becomes clear that trainings on these topics in ULSIT have been conducted, but the respondents themselves have not always participated. More than half of the respondents, 55% (n = 22), have not participated in such trainings so far. The majority of respondents participated in data citation styles training, 32.5% (n = 13). The findings indicate that the respondents show a keen interest in conducting similar trainings in the future. Prevalent among the respondents is the need to have a formal training on: DMP 70% (n = 28), metadata 57.5% (n = 23), version control of data sets 50% (n = 20), consistent file naming 45% (n = 18), and data citation styles 40% (n = 16). Only 3 respondents (7.5%) declared that they are not interested in such training.

4 Overview of the State of Open Access to Research Data in Bulgaria

DL and RDM competency is increasingly important to the academic community in Bulgaria. At a European Union Council meeting on May 27, 2016, EU member states agreed to a goal of making all scientific research papers at state universities within the EU freely available by 2020. Member states and stakeholders were encouraged to set up strategies accordingly and to implement the use of Data Management Plans as a standard scientific practice in their national research programmes. They were invited to contribute to the realization of the Open Science Policy Platform and to the "European Cloud Initiative - Building a competitive data and knowledge economy in Europe" [8].

Bulgaria has a strong background in OA through its participation in the OpenAIRE and OpenAIRE plus projects, each of which provide open access infrastructure to European countries [9]. Since 2005 two associations – National Research Network [10]

and Bulgarian Research and Educational Network [11] – were directed to develop and operate Bulgarian research and educational networks, as well as to facilitate the integration of Bulgarian scientific, educational and cultural resources through a high performance information infrastructure in the global information space.

In spite of these positive steps, Bulgaria is far from achieving the goals set. A range of problems was reported in the "Strategy for Development of Higher Education in the Republic of Bulgaria for the 2014–2020 Period," an analysis of the state of higher education in Bulgaria. Problems reported include:

- difficulties in the quality and compatibility of Bulgarian higher education (HE) in comparison to European HE systems
- weak links between the HE and the needs of business and public institutions
- insufficient links between training and research
- need of improvement in the management system of HE institutions and the HE institutions network
- problems associated with attracting and retaining faculty
- inadequate opportunities for lifelong learning and others [12].

In this context, a serious issue is that due to lack of effective implementation on an national level of RDM policy and regulations – the majority of research data of Bulgarian scientists still are not collected and archived and cannot be accessed.

Nowadays there are six OA Bulgarian repositories, namely:

- Bulgarian OpenAIRE Repository (incl. Bulgarian publications funded by FP7/ERC and EC projects)
- New Bulgarian University Scholar Electronic Repository
- Repository of Institute of Mathematics and Informatics at Bulgarian Academy of Sciences (IMI-BAS)
- Research at Sofia University Repository
- Academic Research Repository of Burgas Free University
- Electronic Repository of the Central Medical Library at Medical University of Sofia.

But, according to "OpenAire in Bulgaria" statistics, in May 2018 there are only 2595 OA publications available from two OpenDOAR repositories: 90 publications from the Bulgarian OpenAIRE Repository and 2507 publications from the Bulgarian Digital Mathematics Library at IMI-BAS [9]. Obviously, there is a need for expansion of efforts in the field of OA and OA to research data in Bulgaria.

Positive developments are expected as a result of three government documents regarding OA and Open Science, accepted in 2017: "National Strategy for Development of Scientific Research in the Republic of Bulgaria 2017–2030: Better Science for Better Bulgaria" (NSDSR), "Bulgaria National Roadmap for Research Infrastructure, 2017–2023", and "Diagnostic Review Mapping of Research Infrastructures and Equipment In Bulgaria" [4, 13, 14]. The first recommendation of the NSDSR is for the Ministry of Education and Science (MES) to establish a policy of OA on the basis of the green model. This should encompass all research institutions that perform and/or disseminate fully or partially state-funded research in order that the results of state-funded research are disseminated to the greatest extent possible.

5 Recommendations for ULSIT Institutional Policy

In this research paper the authors examined the current state of data literacy, data sharing and reuse perceptions and practices among researchers at a single academic institution – University of Library Studies and Information Technologies (ULSIT). Based on the survey results, there are gaps in knowledge that suggest a need for appropriate DL and RDM training at the university. The main findings of the survey, can be summarized as follows: the level of knowledge of respondents regarding DL, RDM awareness and skills are far from satisfactory. The majority of researchers are uncertain about RDM related issues and OA policies and practices. At the same time, the findings indicate a high degree of interest for training on these topics. This interest could be a good basis for planning future learning initiatives. However, one must consider that when planning, organizing and implementing metadata and/or Data Management Plan trainings, trainees cannot be approached as a homogeneous group of knowledge and understanding. The trainings should be selectively offered to different group of learners. Measures should be taken to ensure understanding and implementation of RDM policy in ULSIT.

The analysis of the findings resulted in the creation of a *document with recommendations* to be used as the basis for further reflection on the institutional RDM policy and the improvement of the Quality of Education Management System (QEMS) at ULSIT. Recommendations include the following:

- necessity of implementation of institutional RDM policy connected with open data policy;
- development of institutional repository and research data services for research data storage, analysis and curation as a first step on the Open Access Green Road;
- implementation of DMP in research process, establishment of prescribed metadata set for uploading data into a university repository and of specific guideline for citing data;
- organizing appropriate DL and RDM training, according to international experiences and standards;
- ULSIT must be in collaboration with all stakeholders in Bulgaria involved in the development and management of research networks and with the integration of Bulgarian educational, scientific and cultural resources in the European and world information society.

The strengths of ULSIT in this area are: an established science infrastructure, which includes “John Atanasov” - Digital and 3D national center for science and research in ICT (Living labs, Internet of Things and Smart Cities), LibLab (Laboratory of Library Technologies, Communications and Informing), UNESCO Interfaculty Chair “ICT in Library Studies, Education and Cultural Heritage” and specialized institutes, dedicated to research activities in the field of Bulgarian cultural and historical heritage, working in cooperation with scientific groups from Bulgarian Academy of Science and other organizations. ULSIT strengths also include long-time experienced cooperation with partners from the national system of universities, libraries and museums. ULSIT successfully collaborates with creative and re-creative industries, also.

The systematized conclusions and recommendations from the conducted survey are provided to all individuals and groups responsible for quality standards design at the university. They could be used in the development of a project in the frame of the Operational Programme of EC "Science and Education for Smart Growth 2014–2020" [15]. The implementation of the "Data Literacy Survey" was very useful also in relation to current amendments (May 2018) in the Act for the Development of the Academic Staff in the Republic of Bulgaria [16].

6 Future Plans

Researchers from other universities continue to be interested in the survey. In the period of March–May 2018, the "Data Literacy Survey" was conducted in University of Chemical Technology and Metallurgy in Sofia and in Konstantin Preslavsky University of Shumen. These new sets of data will give authors a chance to make comparisons and to stimulate inter-institutional dialog and collaboration for needed improvements.

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